DHEEMANT JALLEPALLI

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EDUCATION

Carnegie Mellon University (CMU), MS in Biomedical Engineering (Research) — Pittsburgh, PA (2024 – 2025)
GPA: 3.96 / 4.00
Supervisor: Prof. Douglas J. Weber, NeuroMechatronics Lab
Thesis: Gamified Surface Electromyography Neuromotor Interface

Indian Institute of Technology (IIT) Jodhpur, BTech in Bioengineering — Jodhpur, India (2020 – 2024)
Minor: Science of Intelligence
GPA: 7.87/10.00
Supervisor: Prof. Shilpa Dang, Network and Cognitive Neuroscience Lab
Thesis: Assessing the Functional Connective Networks in the Subcortical Area from Human RS-fMRI Data

RESEARCH AND PROFESSIONAL EXPERIENCE

Harvard University, Research Fellow — Cambridge, MA (Mar 2026 – Present)
Supervisor: Prof. Shriya S. Srinivasan

- Designed and conducted a multi-day surface EMG biofeedback pilot study demonstrating that able-bodied participants can learn to volitionally and independently activate two synergistic plantar-flexor muscles, significantly reducing co-contraction and expanding a single degree of freedom into multi-DOF control
- Designed and built a portable multi-channel surface EMG acquisition system with custom analog front-end, filtering, and embedded firmware for real-time biofeedback and gamified motor training, deployed as the at-home platform for an ongoing participant-facing study in lower-limb amputation
- Engineered a compact handheld myoelectric controller mapping muscle activity to standard keyboard input (HID emulation), delivering up to 4 degrees of freedom of control in unmodified commercial games with rapid, unsupervised calibration
- Built a closed-loop functional electrical stimulation (FES) system in a preclinical model, fusing muscle force, intramuscular EMG, and muscle-length measurements to drive stimulation from a muscle-state-aware model within a single real-time control loop
- Designed multi-waveform stimulation protocols to characterize subject-specific fatigue dynamics not captured by any single waveform, generating training data for deep learning controllers that estimate instantaneous force from stimulation history and muscle state
- Benchmarked learned FES controllers against conventional PID control on force-tracking error, overshoot, and time to fatigue during prolonged stimulation

Carnegie Mellon University, Research Engineer — Pittsburgh, PA (Aug 2024 – Mar 2026)
Supervisor: Prof. Douglas J. Weber

- Designed a modular Python biosignal-processing framework to quantify longitudinal motor recovery under spinal cord stimulation in stroke patients from multimodal wearable data, deriving autonomic biomarkers from electrodermal activity decomposition and heart rate variability during affected-limb tasks
- Developed an ML-based neuromotor interface decoding 8-channel surface EMG into continuous wrist and finger movements for real-time gamified human–machine interaction (up to 92% gesture classification accuracy), and integrated it with a custom 3D-printed multi-DOF robotic arm for device control
- Showed that proportional velocity-based continuous control outperformed both position-based and discrete gesture decoding for naturalistic game control, the basis of the MS thesis and a manuscript in preparation
- Built a closed-loop neural stimulation interface on CorTec's implantable device API, implementing parameter validation, frequency-based mapping, and waveform visualization for safe neuromodulation
- Implemented HD-EMG motor unit decomposition post-processing (spatial filtering, recruitment and derecruitment threshold detection, conduction velocity estimation, spike train analysis) within a GUI framework

Carnegie Mellon University, Research Assistant — Pittsburgh, PA
Supervisor: Prof. Sossena Wood — (Jan 2025 – Dec 2025)

- Conducted investigations of cerebral and forearm blood flow in adult sickle cell disease populations, addressing a critical gap left by prior pediatric-focused studies
- Extracted pulsatile hemodynamic features (peak systolic flow, end diastolic flow, pulsatility index, area under the curve) and applied linear regression modeling to evaluate associations with hematocrit and SCD status
- Characterized oxygen extraction fraction and cerebral oxygen metabolism in adult SCD patients to inform disease pathophysiology and potential therapeutic targets

Center for Technology Transfer and Enterprise Creation, CMU — Pittsburgh, PA
Technology Transfer Intern — (May 2025 – Aug 2025)

- Prepared technology assessments for CMU inventions covering prior art, competitive landscape, unmet need, market opportunity, and potential licensees
- Supported the IP and commercialization process, including invention disclosures, patentability assessments, and licensing strategy, and wrote marketing reports communicating value propositions to industry partners

MILA-Québec AI Institute, Research Collaborator — Montréal, Canada
Supervisor: Prof. Samira Ebrahimi Kahou — (May 2024 – Aug 2024)

- Designed and evaluated a mask-guided patch sampling framework using the Segment Anything Model (SAM) to adaptively select diagnostically relevant regions in Whole Slide Images (WSIs) for computational pathology
- Benchmarked Multiple Instance Learning (MIL) aggregators on CAMELYON17, demonstrating a 50% reduction in patch sampling and memory usage while preserving diagnostic accuracy relative to fixed-grid methods
- Achieved state-of-the-art slide-level classification (AUC = 0.97, +4% over baseline) by scaling adaptive sampling

Indian Institute of Technology Jodhpur, Undergraduate Researcher — Jodhpur, India
Supervisor: Prof. Shilpa Dang — (Aug 2023 – Dec 2023)

- Investigated functional connectivity within the human subcortex using resting-state fMRI, identifying distinct functional communities across the thalamus, basal ganglia, and limbic regions
- Developed an end-to-end neuroimaging analysis pipeline incorporating preprocessing, quality control, Louvain and K-means clustering, and graph-theoretical network analysis to quantify cortical convergence and information flow across subcortical structures

University of Manitoba, Mitacs Globalink Research Intern — Winnipeg, Canada
Supervisor: Prof. Veronica Silva — (May 2023 – Aug 2023)

- Collected whole-body kinematics, eye-tracking, and dual-task cost data to examine how distracted walking affects the functional field of view across age groups
- Investigated age-related changes in reliance on vision during walking and in the ability to adapt gait patterns to changes in terrain

ACHIEVEMENTS

- **Outstanding Presentation Award**, CMU Biomedical Engineering Forum (Sep 2025)
  One of 10 awards among ∼200 presented projects (top 5%); $200 award
- **Biomedical Engineering Research Excellence Award**, Carnegie Mellon University (May 2025)
  One of 5 selected by the Graduate Committee for research achievements and academic performance; $6,000 award
- **Biomedical Engineering Department Head's Fellowship**, Carnegie Mellon University (Aug 2024)
  One of 5 fellows selected from an incoming cohort of 28; $5,000 award
- **Mitacs Globalink Research Internship - competitively selected**, Government of Canada (May 2023)
  One of ∼2,000 selected worldwide for Canada's flagship undergraduate research program; $9,000 research fellowship
- **Travel Grant - competitively awarded**, India–EMBO Noninvasive Brain Stimulation (Dec 2022)

- **Machine Learning Summer School Fellowship**, Amazon (Jul 2022 & Aug 2023)
  Among the top 3,000 students selected nationwide in two consecutive editions by entrance examination in ML
- **Chanakya Undergraduate Fellowship - competitively selected**, IIT Jodhpur (Jul 2022)
  One of 3 fellows selected from a cohort of 30 (top 10%); INR 60,000 research fellowship
- **Summer Research Internship Program - competitively selected**, IIT Gandhinagar (May 2022)
  One of 155 interns selected from 24,000+ applicants nationwide (0.6% selection rate)
- **Winner**, RecurDyn Winter Simulation Challenge, IIT Delhi / IIT Jodhpur (Dec 2021 – Feb 2022)
  Led a four-member team to 1st of 11 in a national multibody-dynamics simulation challenge; INR 10,000 prize
- **Stage 1 Full Score (100%)**, ABU Robocon India 2022 (DD-Robocon), IIT Jodhpur team (2022)
  Designed the R2 ball-picking mechanism; design document scored 100% and qualified for the national round

PUBLICATIONS

- **Jallepalli, D.**, & Dang, S. (2025). Cortico-subcortical converging organization at rest. *Scientific Reports*, 15(1), 32133
- Dan, A., Chaudhary, S., **Jallepalli, D.**, Samiur, M., & Gupta, S. (2024). Enhanced Graphical User Interface (GUI) of Recursive Dynamics Simulator (ReDySim) for Multibody Systems. In S. K. Ghoshal, A. K. Samantaray, & S. Bandyopadhyay (Eds.), *Recent Advances in Industrial Machines and Mechanisms (IPROMM 2022)*. Lecture Notes in Mechanical Engineering. Springer, Singapore
- Bouchiat, N., **Jallepalli, D.**, Chandra, S., Chen, K., Leestma, J., Srinivasan, S.S. Fatigue-Aware Closed-Loop Functional Electrical Stimulation Using Muscle-State-Informed Deep Learning Controllers. *Manuscript in preparation*
- **Jallepalli, D.**, Borda, L., Weber, D. Wearable EMG-Based Human–Computer Interface for Gamified Interactions. *Manuscript in preparation*

CONFERENCE PRESENTATIONS AND POSTERS

- **Jallepalli, D.**, Roy, S., Meinert-Spyker, E., Xu, J., Wood, S. *Cerebral and Peripheral Microvascular Function in Sickle Cell Disease: Blood Flow and Oxygen Extraction via Diffuse Correlation and Near-Infrared Spectroscopy.* Poster presented at the 67th American Society of Hematology (ASH) Annual Meeting and Exposition, Orlando, FL, Dec. 2025
- **Jallepalli, D.**, Borda, L., Weber, D. *Gamified Real-Time sEMG Brain–Machine Interface.* Poster presented at the IEEE EMBS 12th International Conference on Neural Engineering, San Diego, CA, Nov. 2025
- **Jallepalli, D.**, Borda, L., Weber, D. *Machine Learning-Driven Myoelectric Control for a Multi-Degree-of-Freedom Upper Limb System.* Poster presented at the 7th Carnegie Mellon Forum on Biomedical Engineering, Pittsburgh, PA, Sep. 2025. **Outstanding Presentation Award**
- Dan, A., Chaudhary, S., **Jallepalli, D.**, Samiur, M., Gupta, S. *Enhanced Graphical User Interface (GUI) of Recursive Dynamics Simulator.* Poster presented at the 2nd International Conference on Industrial Problems on Machines and Mechanism, Dhanbad, India, Dec. 2022

INVITED TALKS

Sep 2026 — *From Debugging Circuits to Debugging the Nervous System*, Alumni Lecture Series, Department of Bioscience and Bioengineering, IIT Jodhpur — Virtual

TEACHING & LEADERSHIP

Volunteer, National Biomechanics Day, *Harvard University* (May 2026)
Led interactive biomechanics demonstrations and STEM outreach for K–12 students.

Teaching Assistant, Introduction to Neural Engineering, *Carnegie Mellon University* (Jul 2025 – Dec 2025)
Mentored students through lab sessions and assignments on neural signal processing and brain–computer interfaces.

Biomedical Engineering Ambassador, *Carnegie Mellon University* (May 2025 – Dec 2025)
Represented the department at outreach events and advised prospective students on academics and research.

Teaching Assistant, Introduction to Biomedical Engineering, *CMU* (Jan 2025 – May 2025)
Supported lab sessions, grading, and office hours on core biomedical engineering principles and techniques.

Teaching Assistant, Introduction to Cognitive Science, *IIT Jodhpur* (Jul 2023 – Dec 2023)
Assisted with lectures, programming assignments, and exams in cognitive science and AI.

Mentor, RecurDyn Winter Simulation Challenge, *IIT Delhi / IIT Jodhpur* (Dec 2022 – Feb 2023)
Mentored the 2022 cohort: designed simulation problem sets, led weekly tutorials, and judged final presentations.

Coordinator, BioX Club for Advanced Biological Systems, *IIT Jodhpur* (Aug 2022 – Jan 2023)
Organized interdisciplinary workshops, research seminars, and technical events in bioengineering.

SKILLS

Programming	Python, MATLAB, C/C++, SQL, LaTeX; PyTorch, TensorFlow, Scikit-learn, SciPy, NumPy, Pandas, OpenCV, MNE-Python, Weights & Biases, Git
Machine Learning	Supervised and deep learning (CNNs, RNNs), time-series and probabilistic modeling, feature engineering, dimensionality reduction, model evaluation
Signal Processing	EEG/EMG/HD-EMG analysis, time–frequency analysis, neural decoding, motor unit decomposition, spike train analysis, EDA/HRV analysis, motion capture (Vicon)
Control & Stimulation	Closed-loop and real-time control, PID design and tuning, functional electrical stimulation (FES), waveform design, force and muscle-length transduction, intramuscular EMG
Hardware & Software Systems	Analog front-end and PCB design, embedded firmware, wearable and research-grade EMG systems, implantable neural interfaces, multi-DOF robotic systems, GUI development (PyQt/PySide, MATLAB App Designer, PyGame), hardware–software integration
Fabrication	3D printing (FDM, SLA, SLS, FRESH), CAD (Fusion 360, SolidWorks)